Gradient flow dynamics of shallow ReLU networks for square loss and orthogonal inputs

Etienne Boursier
EPFL

Loucas Pillaud-Vivien
EPFL

Nicolas Flammarion
EPFL

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Introduction

Mystery of training 2-layer ReLU neural networks for regression with gradient descent

- Do they converge to 0 training loss?
- Do they show some simplicity bias?
- What is the role of initialisation?

→ *partial* answers for large initialisation - classification - infinite width networks

<u>This work</u>: complete answer for orthogonal data and small initialisation (feature learning)
+ complete description of the dynamics

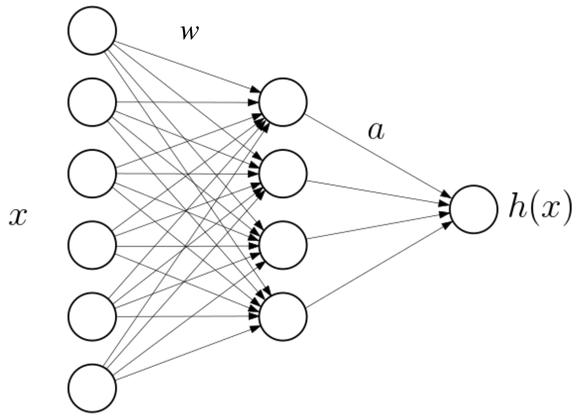
Setting

Given n data points $x_k, y_k \in \mathbb{R}^d \times \mathbb{R}$, minimise square loss over parameters $\theta = (a_j, w_j)_{j\in[m]}$

$$L(\theta) := \frac{1}{2n}\sum_{k=1}^{n}\left(h_\theta(x_k) - y_k\right)^2, \text{ where } h_\theta(x) := \sum_{j=1}^{m} a_j\,\sigma(\langle w_j, x\rangle) \text{ and } \sigma(x) = \max(x,0)$$

Assumption

Inputs are orthonormal: $\forall k, k', \langle x_k, x_{k'}\rangle = \mathbf{1}_{k=k'}$



Gradient flow as continuous approximation of gradient descent

$$\frac{\mathrm{d}\theta^t}{\mathrm{d}t} = -\nabla L(\theta^t) = -\frac{1}{n}\sum_{k=1}^{n}\left(h_{\theta^t}(x_k) - y_k\right)\nabla_\theta h_{\theta^t}(x_k)$$

Initialisation

scale of initialisation

Balanced initialisation:

$$\begin{cases} w_j^0 = \lambda\, g_j \text{ where } g_j \sim \mathcal{N}(0, I_d) \\ a_j^0 = \mathsf{s}_j \|w_j^0\| \text{ where } \mathsf{s}_j \sim \mathcal{U}(\{-1,1\}) \end{cases}$$

Lemma

$\boxed{a_j^t = \mathsf{s}_j \|w_j^t\| \neq 0 \quad \text{for all } t}$ (balancedness)

Define

$$S_+ = \left\{ j \in [m] \mid \mathsf{s}_j = +1 \text{ and for all } k \text{ such that } y_k > 0,\ \langle w_j^0, x_k \rangle \geq 0 \right\}$$

$$S_- = \left\{ j \in [m] \mid \mathsf{s}_j = -1 \text{ and for all } k \text{ such that } y_k < 0,\ \langle w_j^0, x_k \rangle \geq 0 \right\}$$

Assumption

$\boxed{S_+ \text{ and } S_- \text{ are both non-empty}}$

$\rightarrow$ requires $m \gtrsim 2^n$ to hold w.h.p.

Main result

Theorem

If $\lambda \leq \lambda^*$, the gradient flow converges to some θ^*_λ such that $L(\theta^*_\lambda) = 0$. Furthermore,

$$\lim_{\lambda \to 0} \lim_{t \to \infty} \theta^t = \theta^* \in \arg\min_{L(\theta)=0} \|\theta\|^2$$

For small initialisation

- converges to **0 training loss**
- **implicit bias** towards minimal norm interpolator

+ many other insights from proof

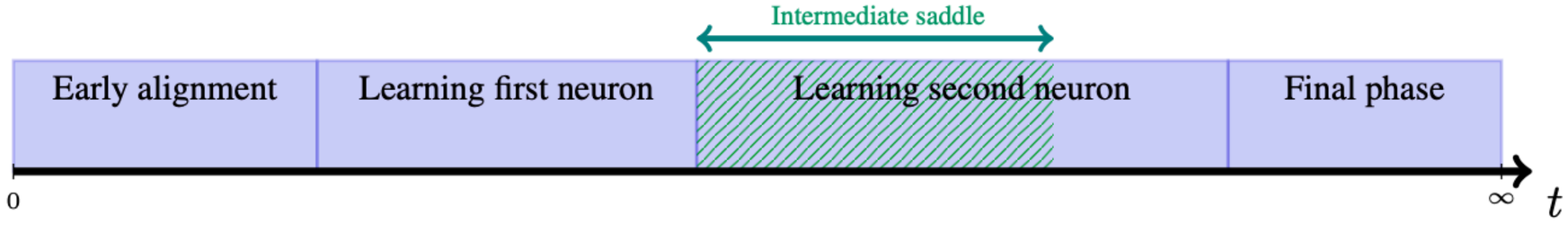
Estimated solution

Minimal norm interpolator

$$h_{\theta^*}(x) = n\langle D_+, x\rangle_+ - n\langle D_-, x\rangle_+ \qquad \text{where} \quad \begin{cases} D_+ = \frac{1}{n}\sum_{k|y_k>0} y_k x_k \\ D_- = \frac{1}{n}\sum_{k|y_k<0} y_k x_k \end{cases}$$

→ equivalent to a 2-neurons non-linear network

Training dynamics



Complete training trajectory can be described

Intermediate saddle point where only one neuron (D_+) has been learnt

Implicit bias

Implicit bias

Minimal ℓ_2 norm parameters $\Longrightarrow$ minimal $\mathscr{F}_1$ norm interpolating function

$\rightarrow$ **Good generalisation** guarantees (see Bach, 2017)

Number of neurons

$m \gtrsim 2^n$ is necessary to reach minimal norm interpolator

Weaker bias (and 0 loss) still happens with less neurons at initialisation

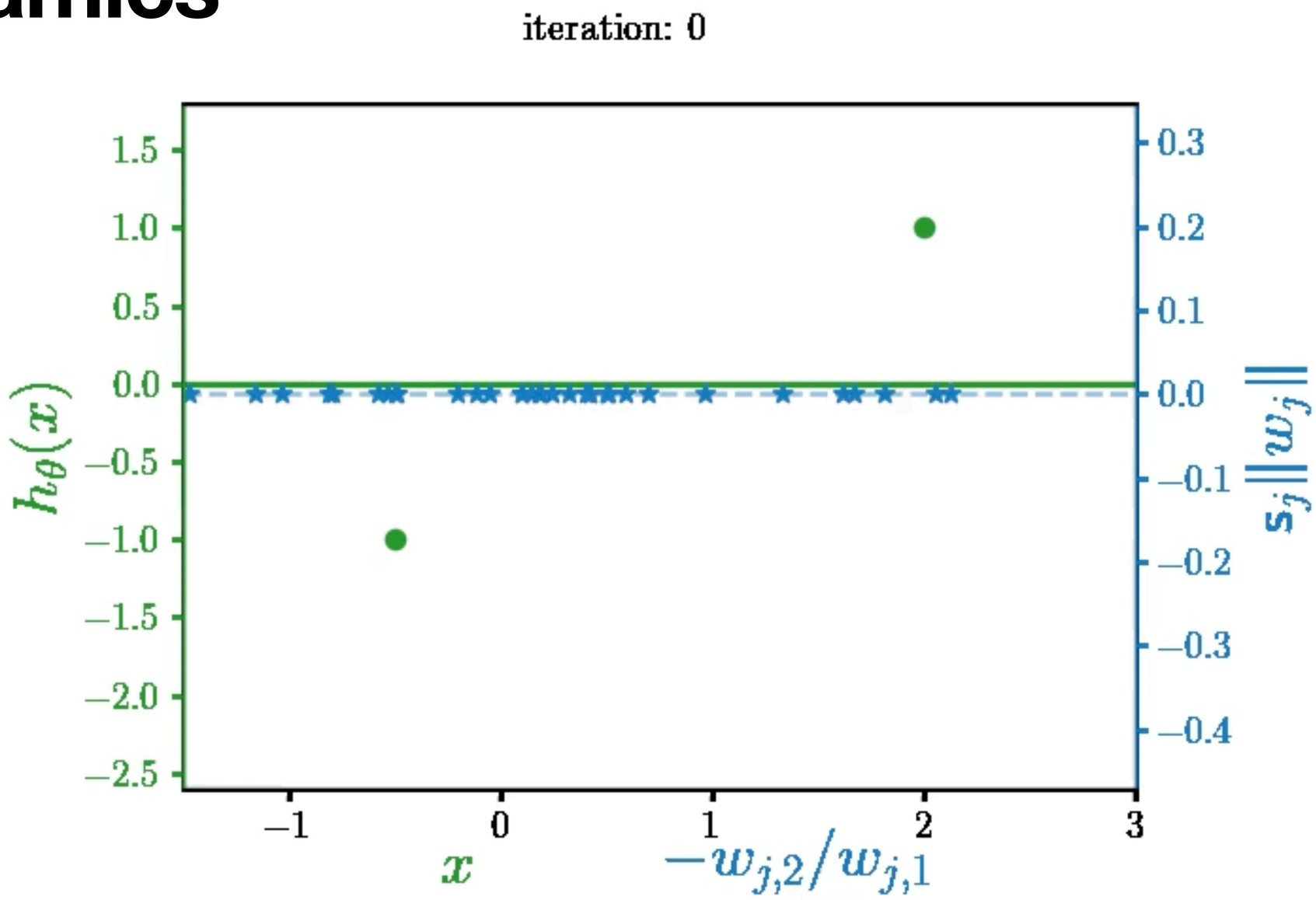
Training dynamics

Orthogonal data with $d = 2$
(bias = second coordinate)

- Dot: data points
- Line: estimated function
- Star: single neuron (w_j, a_j)

  x value $=$ kink position

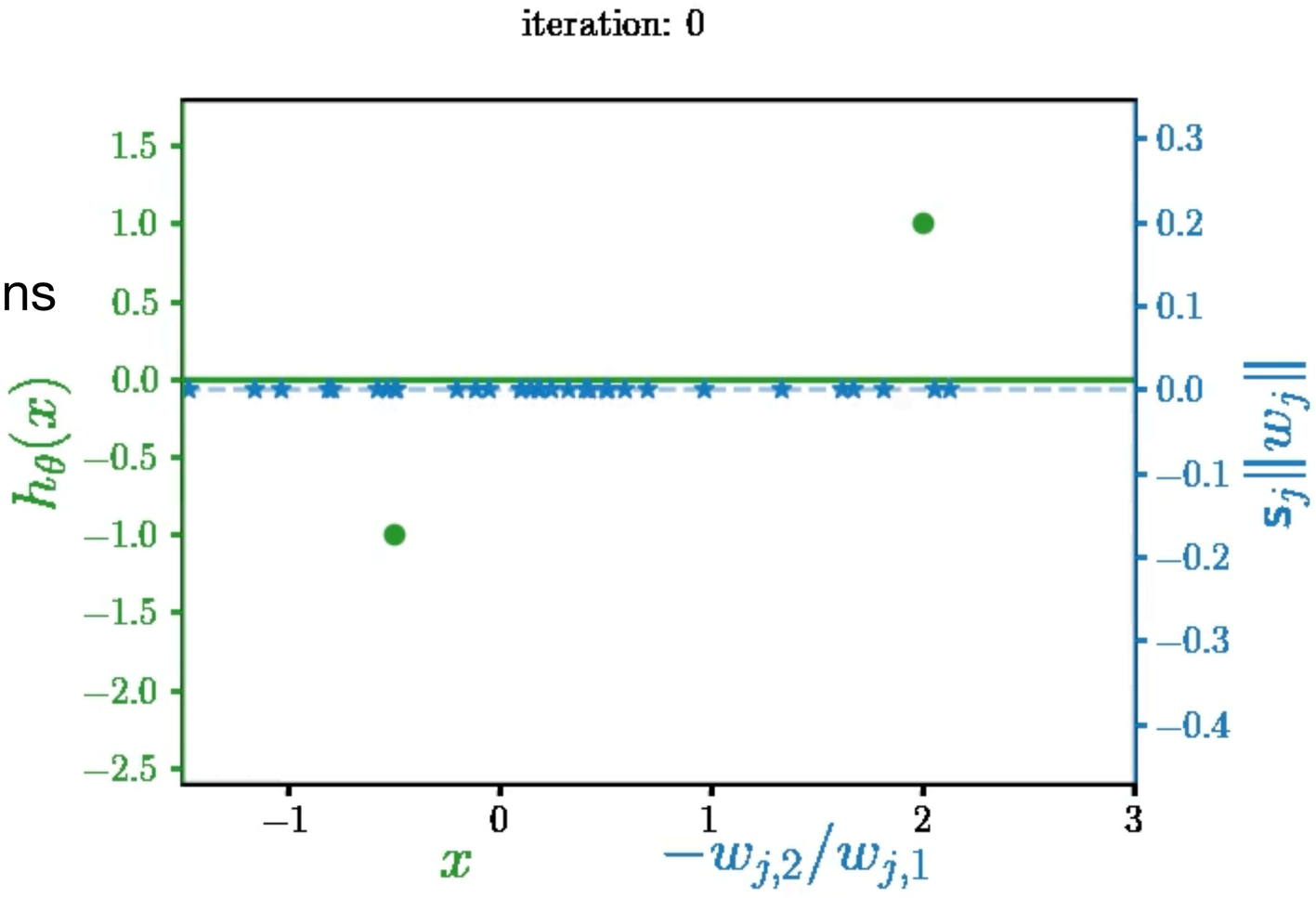
  y value $= a_j$



Phase 1: early alignment

- Neurons remain small in norm
- Neurons align according to key directions (form clusters)

→ also holds in the general case



Phase 1: early alignment

Let $\bar{w}_j = \dfrac{w_j}{\|w_j\|}$, then

$$\begin{cases} \dfrac{\mathrm{d}\bar{w}_j^t}{\mathrm{d}t} = \mathsf{s}_j \left(D_j^{\theta^t} - \langle D_j^{\theta^t}, \bar{w}_j^t \rangle \bar{w}_j^t \right) \\ \dfrac{\mathrm{d}\|w_j^t\|}{\mathrm{d}t} = \mathsf{s}_j \|w_j^t\| \langle D_j^{\theta^t}, \bar{w}_j^t \rangle \end{cases}$$

evolves in time $\mathcal{O}(1)$

evolves in time $\mathcal{O}\left(-\ln(\lambda)\right)$

Assume $\|D_+\| > \|D_-\|$ where $D_\pm = \dfrac{1}{n} \sum_{k=1}^{m} (y_k)_\pm \, x_k$

Lemma

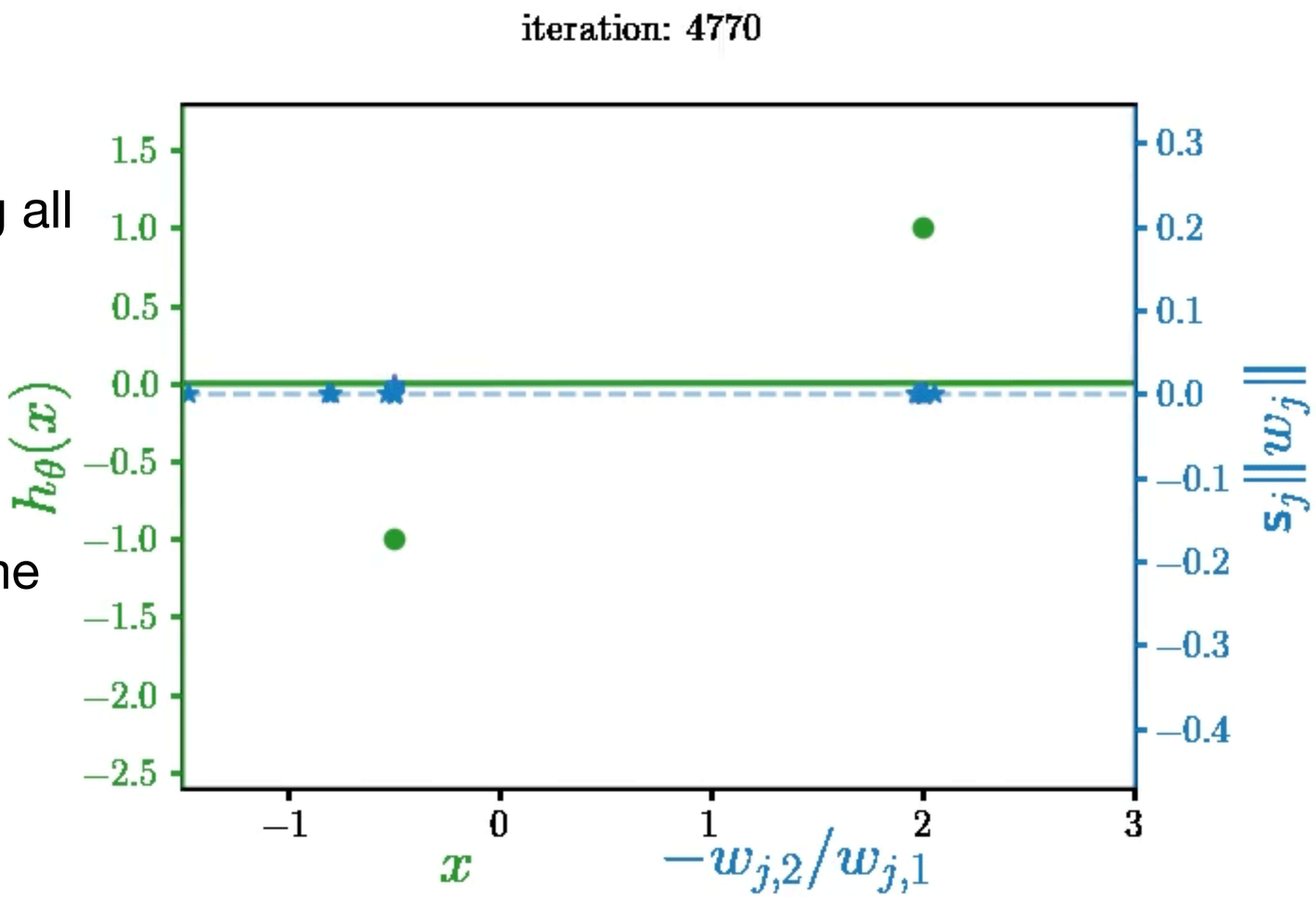
For $\lambda \leq \lambda^*$, we have for $t_1 = \dfrac{-\varepsilon}{\|D_-\|} \ln(\lambda)$:

1. Neurons in $S_\pm$ are aligned: $\forall j \in S_\pm, \langle \bar{w}_j^{t_1}, \pm D_\pm \rangle = \|D_\pm\| + o_\lambda(1)$

2. All neurons have small norm: $\forall j, \|w_j^{t_1}\| = o_\lambda(1)$

Phase 2: fitting positive labels

- A first cluster grows in norm until fitting all positive labels
- They keep a fixed direction while growing
- Other neurons stay idle in the mean time



Phase 2: fitting positive labels

$u_+(t)$ grows exponentially towards $n\|D_+\|$

For $u_+(t) = \sum_{j \in S_+} \|w_j^t\|^2$, we have

$$\frac{\mathrm{d}u_+(t)}{\mathrm{d}t} \approx 2\|D_+\| \left(1 - \frac{u_+(t)}{n\|D_+\|}\right) u_+(t)$$

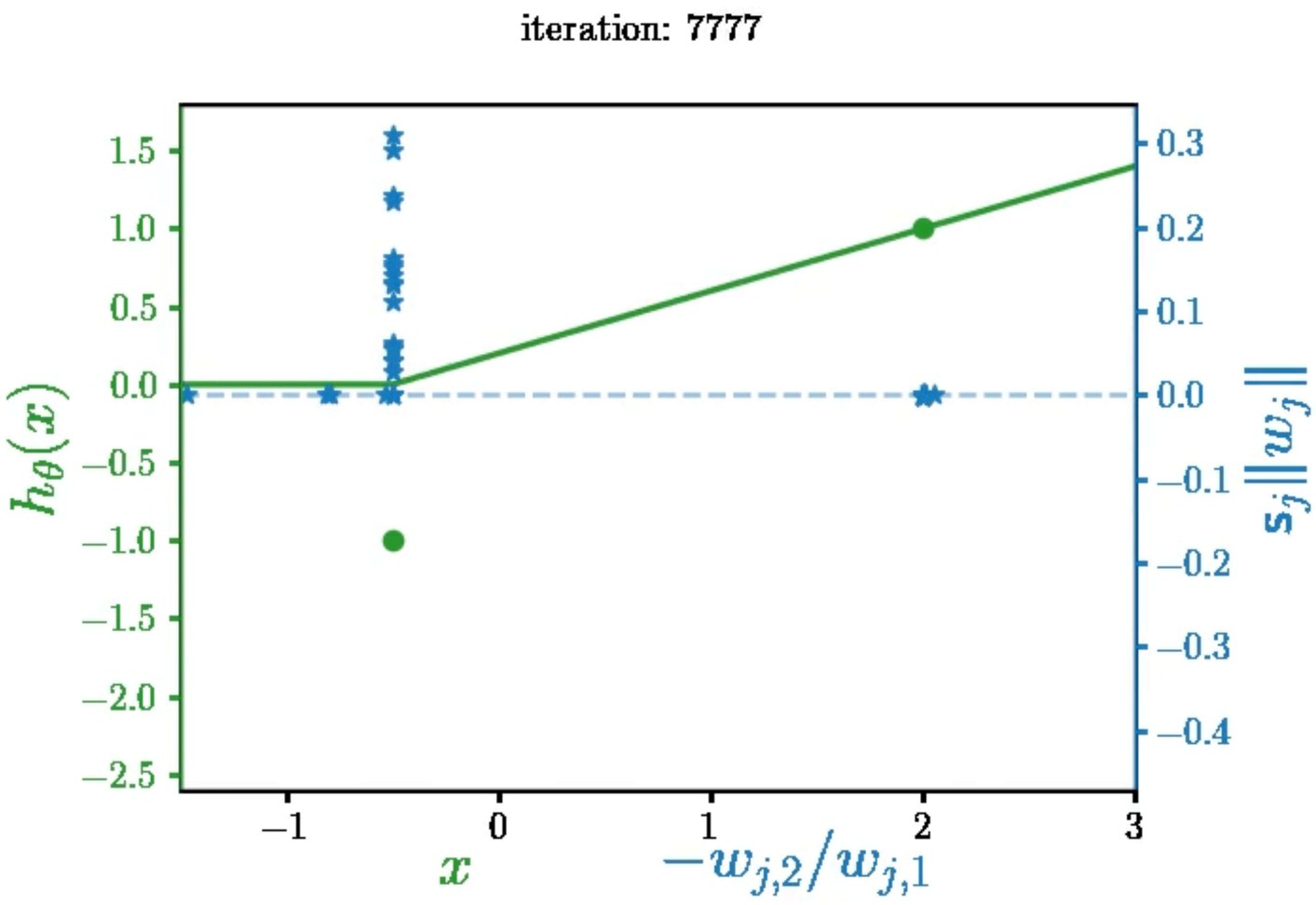
Lemma

For $\lambda \leq \lambda^*$, we have for $t_2 \leq -\frac{1+3\varepsilon}{\|D_+\|}\ln(\lambda)$:

1. Neurons in $S_\pm$ are aligned: $\forall j \in S_\pm, \langle \bar{w}_j^{t_2}, \pm D_\pm \rangle = \|D_\pm\| + o_\lambda(1)$
2. Neurons in S_+ have large norm: $\sum_{j \in S_+} \|w_j^{t_2}\|^2 = n\|D_+\| + o_\lambda(1)$
3. Other neurons have small norm: $\forall j \notin S_+, \|w_j^{t_2}\| = o_\lambda(1)$

Phase 3: fitting negative labels

- The second cluster grows in norm until fitting all negative labels
- They keep a fixed direction while growing
- Other neurons stay idle



Phase 3: fitting negative labels

Similarly to Phase 2 for $r = \dfrac{\|D_+\|}{\|D_-\|}$:

Lemma

For $\lambda \leq \lambda^*$, we have for $t_3 \leq -\dfrac{1+3r\varepsilon}{\|D_-\|}\ln(\lambda)$:

1. Neurons in $S_\pm$ are aligned: $\forall j \in S_\pm, \langle \bar{w}_j^{t_3}, \pm D_\pm \rangle = \|D_\pm\| + o_\lambda(1)$
2. Neurons in S_- have large norm: $\sum_{j \in S_-} \|w_j^{t_3}\|^2 = n\|D_-\| + o_\lambda(1)$
3. Other neurons did not move: $\forall j \notin S_-, \|w_j^{t_2} - w_j^{t_3}\| = o_\lambda(1)$

Final phase: PL convergence

After phase 3, dynamics nearly converged.

From there, we use a **local Polyak-Łojasiewicz** condition

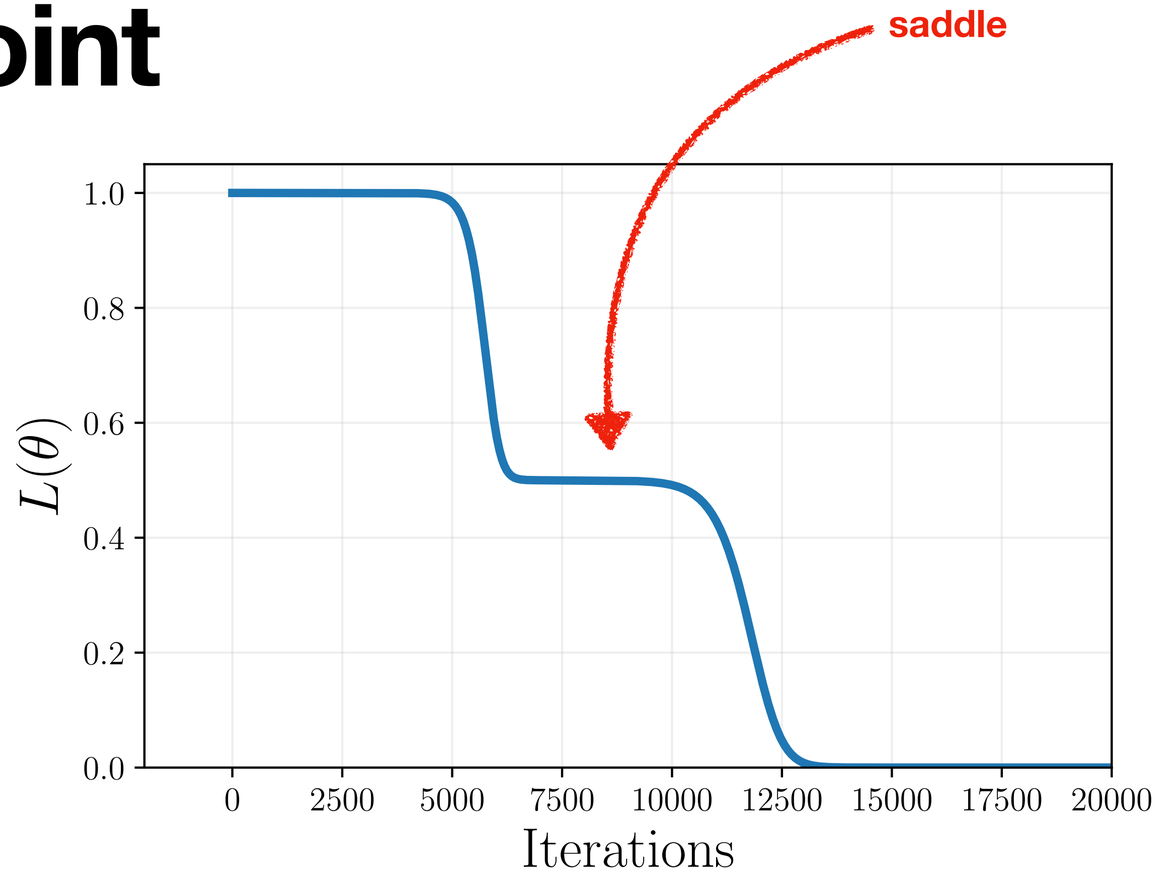
→ parameters stay close from end of Phase 3

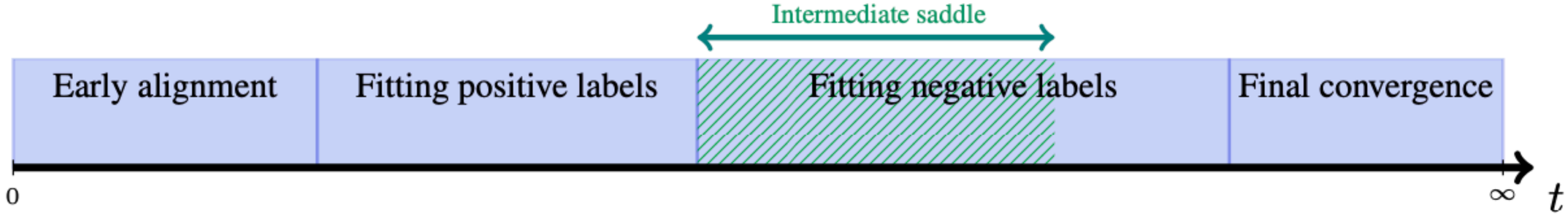
→ convergence towards interpolator at distance $o_\lambda(1)$ of θ^* at exponential rate $\|D_\|$

Intermediate saddle point

At the end of phase 2, goes through a saddle point of the loss.

→ leaves it when norm of second cluster becomes non-zero

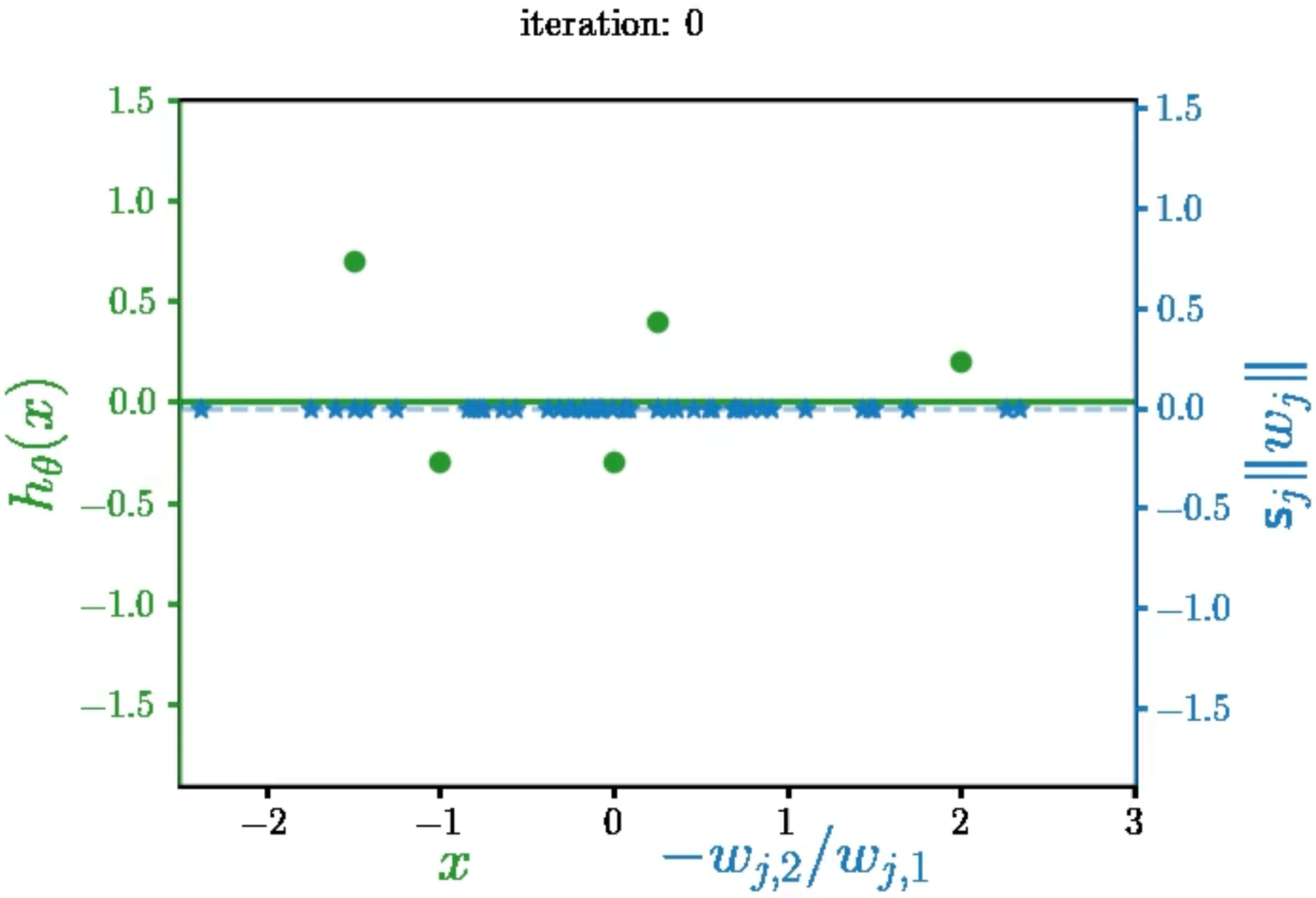




On the general case

Similar phenomena in the general case ($n \geq d$)

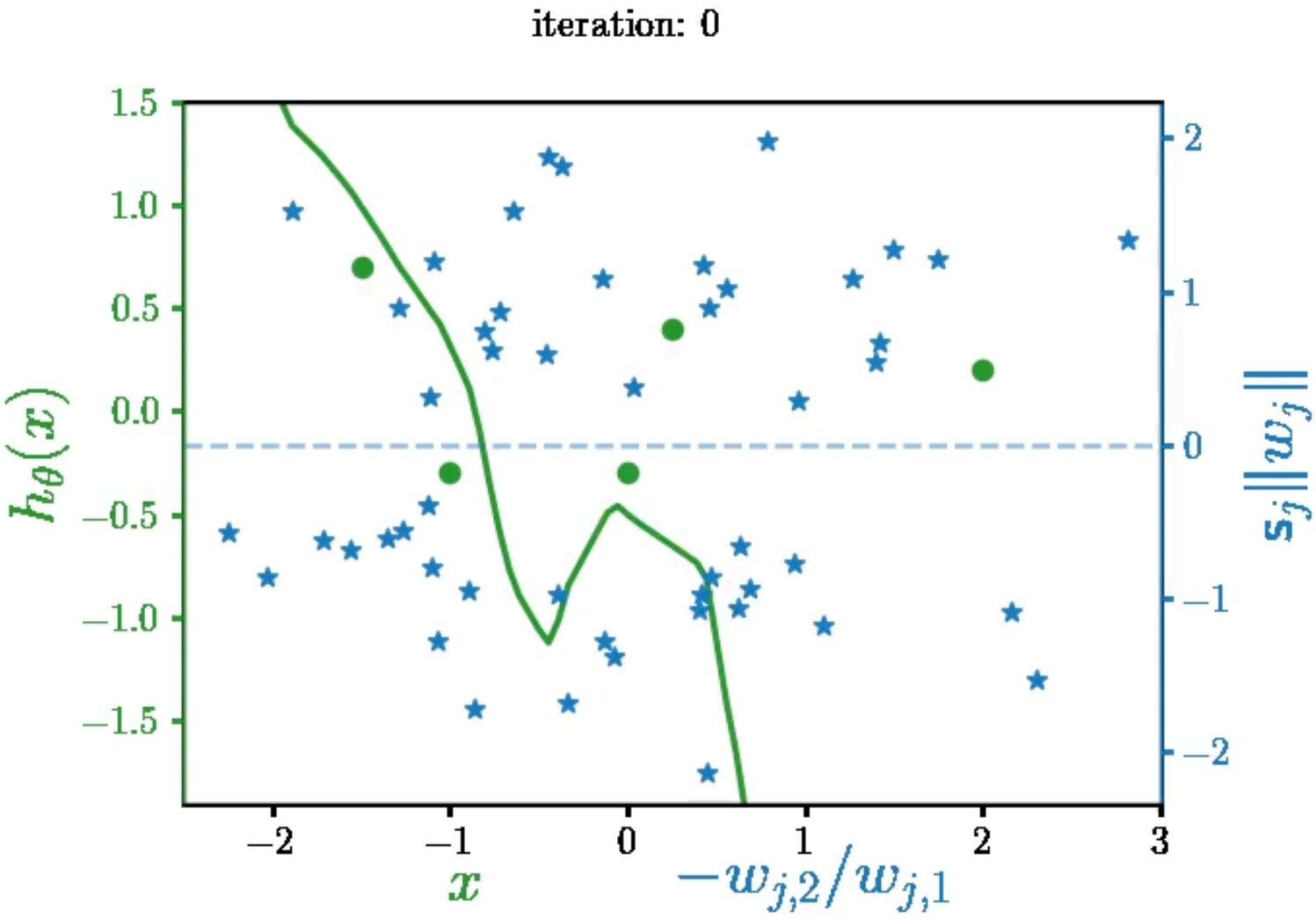
Harder to analyse: neurons can grow in norm and change direction simultaneously



On large initialisation

With larger initialisation

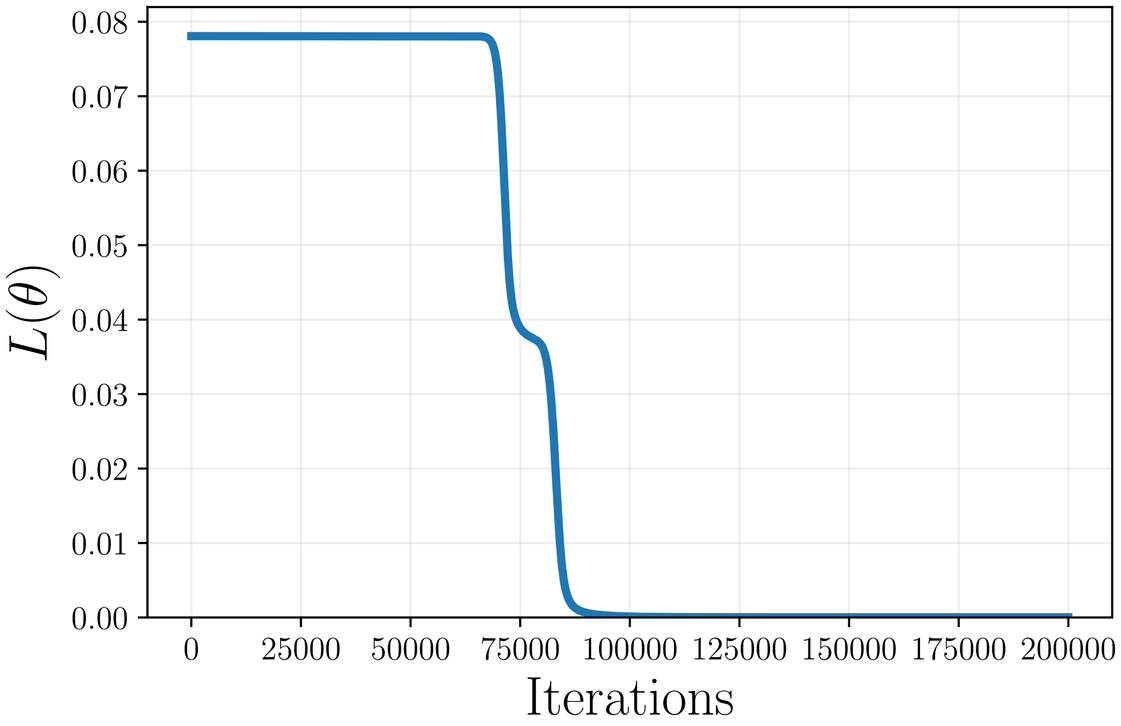
- neurons almost do not move (lazy regime)
- convergence to 0 is faster
- no simplicity bias



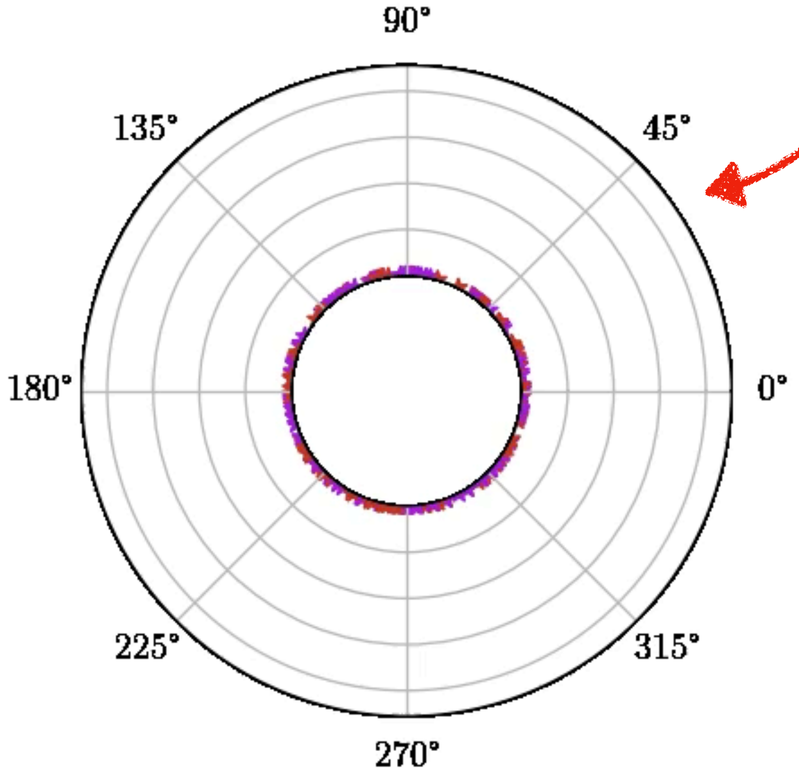
General case: high dimension

Random features with $n = 75$ and $d = 150$ **(near orthogonal data)**

Labels from a random 6 neurons teacher network



Neurons' projection on first 2 principal components



Conclusion

Complete description of training for orthogonal inputs

- Training loss converges to 0
- Implicit bias towards min norm interpolator for small initialisation
- Intermediate saddle point & 3 training phases

General case remains unsolved

→ what insights from orthogonal case?

Thank you!

Code and paper available at eboursier.github.io